Ava Schrandt, Software Engineer | Backend • Data Platforms • AI Systems

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EXPERIENCE

May 2025 — Present

Software Engineer — ClearSky Smart Fleet, Connected Solutions, JLG INDUSTRIES — Hagerstown, MD

- Engineer internal Python/PyQt tooling that automates end-to-end validation of the CS550 ClearSky Smart Fleet device, tracing behavior from embedded firmware and CAN communication through backend services to cloud-reported telemetry.
- Design and maintain time-series telemetry data pipelines using Azure Data Explorer (ADX), developing new feature data flows and troubleshooting production data issues supporting customer-facing ClearSky Smart Fleet applications.
- Develop an internal AI assistant over enterprise data using Microsoft AI Foundry, implementing agent orchestration, retrieval, and validation with Python/FastAPI, TypeScript/React, and vector databases.
- Integrate testing infrastructure with Java backend microservices and package the tooling as a Windows executable for use across internal engineering and validation workflows.

May 2024 — May 2025

Software Engineer Intern — ClearSky Smart Fleet, JLG INDUSTRIES

- Initiated development of an automated end-to-end testing framework for the CS550 ClearSky Smart Fleet device, building Python/PyQt tooling to parse live CAN data and support device-to-cloud validation.
- Enhanced existing internal engineering tools with secure Azure authentication and automated application setup and deployment workflows.

TECHNICAL SKILLS

Python | Java | C++ | C | SQL | TypeScript | JavaScript | FastAPI | REST APIs | Java microservices | Kubernetes | Microsoft Azure | Azure Data Explorer (ADX) | NoSQL | MongoDB | Microsoft AI Foundry | Agent Orchestration | Automated end-to-end testing | PyQt | CAN bus | Windows application packaging | Git | Agile/Scrum | Jira | Confluence | Linux/Unix | Vector Databases | Jama Software | OpenAI API

EDUCATION

Present

B.S. Computer Science — Software Engineering — West Lafayette, IN

PURDUE UNIVERSITY

GPA 3.62 • Dean’s List: Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025

SELECTED PROJECTS

May 2025

SafeEats

Built an iOS application for finding nearby dining options based on dietary restrictions using Python, FastAPI, MongoDB, Google Places, and Swift.

May 2024

Unix Shell Interpreter

Implemented shell behavior and command processing in C/C++ using Lex and Yacc.

May 2024

Serenova Sleep Tracker

Scrum Master for an Agile iOS team; developed features with SwiftUI, Firebase, and Apple HealthKit.